

Auteur: Stan Van Campenhout

Studierichting: Informatica

Oefeningenreeks 6D nr. 30.16

$$\begin{aligned}
 & \sum_{n=2}^{\infty} e^{-\frac{\sqrt{n}}{2}} \\
 & \int_0^{+\infty} e^{-\frac{\sqrt{n}}{2}} dn \stackrel{t^2 \equiv n}{=} 2 \int_0^{+\infty} te^{-\frac{t}{2}} dt = -4 \int_0^{+\infty} td \left(e^{-\frac{t}{2}} \right) \\
 & = \left[-4te^{-\frac{t}{2}} \right]_0^{+\infty} - 8 \int_0^{+\infty} e^{-\frac{t}{2}} d \left(-\frac{t}{2} \right) \\
 & = \left[-4te^{-\frac{t}{2}} - 8e^{-\frac{t}{2}} \right]_0^{+\infty} \stackrel{t=\sqrt{n}}{=} \left[-4\sqrt{n}e^{-\frac{\sqrt{n}}{2}} - 8e^{-\frac{\sqrt{n}}{2}} \right]_0^{+\infty} = \left[\frac{-4(\sqrt{n}+2)}{e^{\frac{\sqrt{n}}{2}}} \right]_0^{+\infty} \\
 & \lim_{a \rightarrow +\infty} \left(\frac{-4(\sqrt{a}+2)}{e^{\frac{\sqrt{a}}{2}}} \right) - \left(\frac{-4(\sqrt{0}+2)}{e^{\frac{\sqrt{0}}{2}}} \right) \\
 & = 0 - (-8) = 8 \Rightarrow \text{bestaat} \\
 & \text{integraalcriterium} \Rightarrow \text{convergent}
 \end{aligned}$$

References